



POLITECNICO DI TORINO

Corso di Laurea in Cybersecurity

Tesi di Laurea

Iterative Approach for Network Security Automation

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Capitolo 1

Introduction

Capitolo 2

Background

2.1 Context

Network security is a critical concern for virtually every organization, from small businesses to large enterprises and public institutions. All of them rely on computer networks to support their operations, and all of them are exposed to the risk of unauthorized access, data breaches, and service disruptions. As a result, enforcing a well-defined and correct security policy is not optional, but instead is a fundamental operational requirement.

Modern enterprise networks are characterized by growing complexity: hundreds of interconnected nodes, heterogeneous services, and security policies that must be consistently enforced across distributed infrastructure. Moreover, network topologies are not static: new services are deployed, nodes are added or removed, and connectivity requirements evolve continuously, making security management an ongoing and demanding task. The cost of managing network security in this context is significant, as it requires specialized expertise, advanced planning, and constant monitoring.

A central challenge in this landscape stems from the evolution of network security architectures. Traditionally, a single perimeter firewall was sufficient to protect a network by controlling traffic at its boundary. However, as networks grew in scale and complexity, this model proved inadequate: a single point of control cannot enforce fine-grained policies across a heterogeneous internal topology. Modern networks have therefore evolved toward microsegmentation, deploying multiple packet-filtering firewall instances strategically distributed across the topology to enforce security policies at a much more granular level. This distributed approach offers clear advantages such as enabling finer-grained traffic control, reduces single points of failure, and allows security policies to be enforced closer to the traffic sources, but it also introduces considerable management complexity. Each firewall instance must be individually configured with a set of filtering rules, and those rules must collectively enforce the intended security policy across the entire network, without conflicts or gaps.

In this context, the configuration of firewalls becomes a critical task. On one hand, overly permissive rules may allow unauthorized traffic to reach sensitive resources, exposing the network to attacks. On the other hand, overly restrictive rules may block legitimate traffic, degrading service availability and causing operational disruptions. Both failure modes carry significant consequences, and yet both are common outcomes of manual configuration process which is inherently prone to errors especially for large-scale networks.

The gap between the complexity of modern networks and the limitations of manual security management calls for automated approaches that can generate correct and optimal firewall configurations without human intervention. It is in this context that VEREFoo was developed: a tool for the automated, formally correct allocation and configuration of distributed firewalls, designed to bridge exactly this gap.

2.2 VEREFOO

2.2.1 Overview

VEREFOO (VERification REasoning on Firewall Optimization) is a tool designed to automate the allocation and configuration of packet-filtering firewalls in a network. It takes as input a network topology, a set of Network Security Requirements (NSRs) and a Firewall Allocation Schema (FAS), which defines the candidate positions in the topology where firewalls may be placed. From these inputs, VEREFOO produces two outputs: an updated topology indicating which nodes have been allocated as active firewalls, and a vendor-independent configuration for each firewall specifying the filtering rules to be enforced. The tool guarantees three fundamental properties: *automation*, as the entire allocation and configuration process requires no manual intervention; *formal correctness*, as the generated configuration is proven to satisfy all specified NSRs; and *optimality*, as the number of allocated firewalls and generated rules is minimised. These guarantees are achieved by encoding the problem as a MaxSMT instance and solving it with the Z3 solver. The resulting configuration can further be translated into vendor-specific formats.

2.2.2 React-VEREFOO

In operational environments, network security policies are not static: NSRs evolve over time as new services are deployed, topology changes occur, or threat models are updated. A fundamental limitation of the monolithic approach used by VEREFOO is that any change to the NSR set triggers a full recomputation of the firewall configuration from scratch. React-VEREFOO addresses this by offering a more efficient approach in the dynamic context described above, avoiding the need to discard a valid existing configuration when only a small portion of the requirements has changed.

React-VEREFOO takes as input both the existing firewall configuration and the updated set of security requirements, and computes only the minimal changes needed to bring the network into compliance. Rather than reconsidering the entire problem, it identifies which requirements are new, which have been removed, and which remain unchanged. Only the newly added requirements drive the reconfiguration: the approach determines which parts of the network are affected and restricts the solver to those areas, leaving the rest of the configuration untouched. The result is an updated firewall allocation and rule set that satisfies all current requirements, with the same guarantees of correctness and optimality as VEREFOO.

Experimental results show that React-VEREFOO is significantly more efficient than vanilla VEREFOO when the proportion of modified NSRs is small, as the reduced problem size leads to much shorter solving times. However, when a large fraction of the requirements changes, the benefits is reduced and React-VEREFOO's performance becomes comparable to that of the monolithic approach. This observation motivates the approach investigated in this thesis: by pushing the incremental logic to its extreme - introducing only one new NSR per iteration - the proportion of modified requirements is always minimised, potentially achieving consistent speedups regardless of the total number of requirements.

Capitolo 3

Thesis Objectives

3.1 Idea of the Solution

As discussed in the introduction, React-VEREFOO is most effective when the number of additional requirements is small relative to the total. The key idea of the approach is to push that observation to its logical extreme: instead of processing all new NSRs at once, the proposed approach introduces them one at a time, turning the configuration problem into a sequence of minimal reconfiguration steps.

The process works as follows. At each iteration, **the current firewall configuration is treated as the initial state and one single new NSR is added as the target**. React-VEREFOO is then invoked to compute the minimal update needed to satisfy that one additional requirement. The output of each iteration, meaning the updated allocation scheme and the associated filtering rules becomes the initial state for the next one. This continues until all NSRs have been incorporated.

Formally, if the set of additional NSRs to be processed is $\{A, B, C, \dots\}$ and P_0 denotes the initial set of properties already enforced in the network, the sequence of iterations is:

1. initial: P_0 , additional: $\{A\}$
2. initial: $\{P_0, A\}$, additional: $\{B\}$
3. initial: $\{P_0, A, B\}$, additional: $\{C\}$
4. ...

Since each invocation of React-VEREFOO deals with exactly one added requirement, the portion of the network that must be reconsidered at each step is minimised by construction. This keeps each individual MaxSMT subproblem small, regardless of the total number of NSRs, with the expectation of reducing overall computation time. The number of iteration steps is equal to the number of new NSRs, which is a linear factor, but the reduced complexity of each step is expected to more than compensate for that.

Capitolo 4

The Iterative VEREFOO Approach

Capitolo 5

Implementation and Validation

5.1 Implementation

5.2 Testing

5.2.1 Test Presentation

5.2.2 Test Configurations

5.2.3 Results

5.2.4 Comparison and Discussion

Capitolo 6

Conclusion and Future Work